

Shirin Hajahmadi

Position: Postdoctoral Researcher, Department of Computer Science and Engineering, University of Bologna
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Web: Personal Website | University Profile



Profile Summary

Postdoctoral researcher specializing in Human-Computer Interaction and Cognitive Augmentation through XR, AI, and Sensors. Proven research record with publications in IEEE VR, ISMAR, and ACM venues; recipient of multiple fellowships and grants; experienced in leading interdisciplinary teams and developing XR-based cognitive enhancement systems. Experienced in designing inclusive XR applications for neurodiverse and physically diverse users.

Research Interests

- Human-Computer Interaction with a focus on immersive technologies (XR, VR/AR/MR) for cognitive augmentation, learning, and collaborative experiences.
- Human-AI interaction and intelligent systems, with emphasis on deep learning for computer vision, large language models (LLMs), and adaptive user interfaces.
- Security and privacy in extended reality environments, focusing on digital identity protection and ethical human-AI interaction.
- Inclusive and diversity-aware design of interactive systems, advancing accessibility and education for neurodiverse and physically diverse users.
- Interdisciplinary approaches bridging computer science, cognitive science, and design, contributing to human-centered digital futures.

Education

University of Bologna, Department of Computer Science and Engineering / VARLab	<i>Bologna, Italy</i>
Ph.D. in Computer Science	<i>Nov. 2021 – Apr. 2025</i>
• Thesis: Cognitive Augmentation Systems through Extended Reality, Artificial Intelligence, and Sensors: A Human-Computer Interaction Perspective	
Institute for Cognitive Science Studies / Shahid Beheshti University	<i>Tehran, Iran</i>
M.Sc. in Cognitive Science (Cognitive Psychology)	<i>Oct. 2016 – Jan. 2019</i>
• Thesis: Map-based alignment effects on virtual spatial navigation in complex curved layouts	
Shahid Bahonar University	<i>Kerman, Iran</i>
B.Sc. in Computer Science	<i>Feb. 2006 – Apr. 2010</i>
• Thesis: Toward the design and development of lexical and syntactic compilers	

Academic Positions

Department of Computer Science and Engineering, University of Bologna	<i>Bologna, Italy</i>
Postdoctoral Researcher	<i>Apr. 2025 – Present</i>
• Conducted research on XR security and privacy paradigms; responsible for design, evaluation, and system development	

Department of Informatics, University of Hamburg

Hamburg, Germany

Visiting Researcher

Sep. 2023 – Nov. 2023

- Collaborated on the design and development of inclusive VR for users with limb loss, and supervised the design and development of an MR assistant for puzzle-solving; conducted the user experience study.

University of Bologna & Anastasis Srl — COMPRENDO Project

Bologna, Italy

Research Fellow

Mar. 2020 – Oct. 2021

- Collaborated on the design and development of inclusive AR and desktop applications aimed at cognitive augmentation for children with autism.

Shahid Beheshti University

Tehran, Iran

Research Assistant

Feb. 2019 – Mar. 2020

- Conducted studies on spatial cognition in virtual environments; conducted user study.

Fellowships & Grants

European Union – NextGenerationEU, University of Bologna

Bologna, Italy

PNRR Research Fellowship (Postdoctoral)

2025 – ongoing

- Research on the design, analysis, and implementation of security and privacy paradigms in extended reality (XR) scenarios.

University of Bologna

Hamburg, Germany / Bologna, Italy

Marco Polo Incentive Fellowship

2023–2024

- International research stay at the University of Hamburg (Department of Informatics).

University of Bologna, Department of Computer Science and Engineering / VARLab

Bologna, Italy

Ph.D. Scholarship

2021–2024

- Doctoral research in Computer Science with a focus on cognitive augmentation through XR, AI, and sensors.

Institute of Advanced Studies (ISA), University of Bologna

Bologna, Italy

ISA Fellowship (International Ph.D. College)

2021–2024

- Interdisciplinary research collaborations and community engagement.

Emilia-Romagna Region

Bologna, Italy

Research Fellowship

2020–2021

- Research on the design, development, and implementation of Augmented Reality and desktop applications for inclusive education.

Cognitive Sciences and Technologies Council, Vice Presidency of Iran

Tehran, Iran

Workshop Grant

2019

- Organizing and leading a workshop series on task design in spatial cognition studies with VR.

Research Collaborations / Partners

Research visit on XR interaction: Collaboration on inclusive VR for users with limb loss and an MR assistant for puzzle-solving project.

Hamburg, Germany

University of Hamburg, Department of Informatics

2024

Collaboration on the Development of an AR + ChatGPT mobile application for inclusive language learning.

Tarragona, Spain

Universitat Rovira i Virgili, Department of Romance Philology

2024

Collaboration on the Sora project: Investigating audiovisual elements and realism in generative AI videos.

Madrid, Spain

Universidad Carlos III de Madrid (Spain)

2024

Teaching & Supervision

eXRAI: Exploring XR and AI in the Creative Industries Workshop (sponsored by University of Bologna and Universidad Rey Juan Carlos)

Bologna, Italy

Workshop Lecturer

Jul. 2024

- Taught Unity for creating virtual environments and supervised the implementation of an example Unity project (6 hours).

University of Hamburg, Department of Informatics

Hamburg, Germany

Mentor (Bachelor Students)

2023–2024

- Supervised group research on social dynamics and performance in mixed reality puzzle activities using different types of virtual assistants.

University of Bologna, Department of Computer Science and Department of Fashion Studies

Bologna / Rimini, Italy

Supervisor (Master Theses)

2022–2023

- Maryam Arjomandi – *VR fashion experience inspired by “Alice in Wonderland”*.
- Igor Iurevici – *User enactments in VR: Smart home interaction study*.
- Ranim Elnabouche – *Virtual fashion show as a gateway to avatar customization*.
- Yue Chen – *Retail–metaverse synergy for consumer engagement*.

Virtual Reality and Augmented Reality Laboratory, MSc Computer Science

Bologna, Italy

Lecturer

2019–2025

- Delivered 11 hours of instruction on immersive system design and evaluation.

Data Science and Immersive Technologies for Fashion E-commerce, MSc Fashion Studies

Rimini, Italy

Lecturer

2020–2024

- Delivered 10 hours of instruction on XR and data-driven fashion retail.

Advanced Information and Communication Technologies, MSc Fashion Studies

Rimini, Italy

Lecturer

2023–2024

- Delivered 18 hours of instruction on advanced ICT methods in fashion studies.

Academic Service

- Ad-hoc Reviewer, Taylor & Francis Journals (e.g., Digital Creativity), 2025
- Ad-hoc Reviewer, Frontiers in Computer Science – Section: Human–Media Interaction, 2025
- Program Reviewer, ACM Symposium on Virtual Reality Software and Technology (VRST), 2025
- Ad-hoc Reviewer, Workshops at IEEE VR, ISMAR, and AIXVR (multiple years, 2023–2025)
- Workshop Organizer:
- Workshop Organizer:

Talks

- **IEEE VR 2025 – The 32nd IEEE Conference on Virtual Reality and 3D User Interfaces:** Investigating the Impact of Voice-only and Embodied Conversational Virtual Agents on Mixed Reality Puzzle Solving (Saint-Malo, France, March 2025).
- **IEEE AIXVR 2025 – XRIM Workshop:** Redefining Fashion: A VR Journey from 360° Runway Show to Contextual Avatar Customization (Lisbon, Portugal, January 2025).
- **CREA(I) 2024:** The influence of audiovisual elements on the realism of generative AI videos: The case of Sora (Santiago de Compostela, Spain, October 2024).
- **GRLMC Seminar 2023:** ARELE-bot: Inclusive ELE Learning with AR and ChatGPT (Tarragona, Spain, December 2023).
- **Digital Research in the Humanities and Arts (DRHA) 2023:** Exploring Symbolic Narratives in Virtual Spaces (Turin, Italy, September 2023).
- **Human-Computer Interaction International (HCII) 2023:** Toward a Shared Experience of Uncertainty of Interpersonal Communication Through an Immersive VR Serious Game (Copenhagen, Denmark, July 2023).

JOURNAL PAPERS

- **Hajahmadi, S.**, Calvi, I., Stacchiotti, E., Cascarano, P., & Marfia, G. (2024). *Heritage Elements and Artificial Intelligence as Storytelling Tools for Virtual Retail Environments*. Digital Applications in Archaeology and Cultural Heritage, e00368.
- **Hajahmadi, S.**, & Marfia, G. (2023). *Effects of the uncertainty of interpersonal communications on behavioral responses of the participants in an immersive virtual reality experience: A usability study*. Sensors, 23(4), 2148.
- **Hajahmadi, S.**, Ghasempouri, S., & Marfia, G. (2023). *Exploring symbolic narratives in virtual spaces: Leveraging curiosity-driven design and the attention-value model for educational VR museum experiences*. Mimesis Journal, 13(2), DRHA Proceedings.

CONFERENCE PAPERS

- **Hajahmadi, S.**, Cascarano, P., Mostajeran, F., Heuer, K., Lux, A., Mends-Cole, G. O., Steinicke, F., & Marfia, G. (2025). Investigating the impact of voice-only and embodied conversational virtual agents on mixed reality puzzle solving. To appear in Proceedings of the 2025 IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR 2025).
- **Hajahmadi, S.**, Stacchio, L., Giacché, A., Cascarano, P., & Marfia, G. (2024, October). Investigating eXtended Reality-powered Digital Twins for sequential instruction learning: the case of the Rubik's cube. In 2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR) (pp. 259-268). IEEE.
- Stacchio, L., Armandi, V., **Hajahmadi, S.**, Donatiello, L., & Marfia, G. (2024). *M-AGEW: Empowering Outdoor Workouts with Data-Driven Augmented Reality Assistance*. In 2024 IEEE International Conference on Artificial Intelligence and eXtended and Virtual Reality (AIxVR) (pp. 301-305). IEEE.
- Sanchez-Acedo, A., Carbonell-Alcocer, A., Cascarano, P., **Hajahmadi, S.**, Vallasciani, G., Gertrudix, M., & Marfia, G. (2024). The influence of audiovisual elements on the realism of generative AI videos: the case of Sora. In Proceedings of the International Workshop on Artificial Intelligence and Creativity (CREAI), co-located with ECAI 2024 (pp. 1-12).
- **Hajahmadi, S.**, & Marfia, G. (2023, July). Toward a shared experience of uncertainty of interpersonal communication through an immersive virtual reality serious game. In International Conference on Human-Computer Interaction (pp. 566-580). Cham: Springer Nature Switzerland.

0.1 Workshop Papers

- **Hajahmadi, S.**, Nabouche, R., Cascarano, P., & Marfia, G. (2025). Redefining fashion: A VR journey from 360° runway show to contextual avatar customization. To appear in Proceedings of the XRiM Workshop at IEEE AIxVR 2025.
- Cascarano, P., Loretto, A., Di Pasquale, A., **Hajahmadi, S.**, Vallasciani, G., Zanuttini, L., Martinoni, M., & Marfia, G. (2025). Diegetic user interfaces in extended reality for 3D medical visualization. To appear in Proceedings of the Healthcare Workshop at IEEE AIxVR 2025.
- Garzarella, S., Vallasciani, G., Cascarano, P., **Hajahmadi, S.**, Cervellati, E., & Marfia, G. (2025). An extended reality platform powered by large language models: A case study on teaching dance costumes. To appear in Proceedings of the Generative AI Workshop at IEEE AIxVR 2025.
- **Hajahmadi, S.**, Clementi, L., López, M. D. J., & Marfia, G. (2024). *ARELE-bot: Inclusive Learning of Spanish as a Foreign Language Through a Mobile App Integrating Augmented Reality and ChatGPT*. In 2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW) (pp. 335-340). IEEE.
- Stacchio, L., Angeli, A., **Hajahmadi, S.**, & Marfia, G. (2021, October). *Revive family photo albums through a collaborative environment exploiting the HoloLens 2*. In 2021 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct) (pp. 378-383). IEEE.
- Stacchio, L., **Hajahmadi, S.**, & Marfia, G. (2021, March). *Preserving family album photos with the HoloLens 2*. In 2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW) (pp. 643-644). IEEE.

POSTER PAPERS

- Cascarano, P., Meglialdi, J., Vallasciani, G., Armandi, V., Augello, G., Carradori, S., **Hajahmadi, S.**, & Marfia, G. (2025). A comparative analysis of 3D modeling methods for integration into an extended reality platform. To appear in Proceedings of the 2nd CFP Posters & Demos Session at IEEE AIxVR 2025.
- **Hajahmadi, S.**, Mostajeran, F., Heuer, K., Lux, A., Mends-Cole, G. O., Cascarano, P., Steinicke, F., & Marfia, G. (2024). *PuzzleAide: Comparing audio and embodied assistants for MR puzzle-solving*. In Proceedings of the ACM Symposium on Spatial User Interaction (SUI).
- Di Maria, G., **Hajahmadi, S.**, Sapienza, S., Marfia, G., & Palmirani, M. (2024). XR4LAW: Implementing an immersive ergonomic user interface for legislative and deliberative institutions. In *Proceedings of the ACM Symposium on Spatial User Interaction (SUI)*.

Skills

RESEARCH & METHODS

- Design and execution of qualitative and quantitative user studies, usability testing, interviews, and surveys.
- Experience with cognitive measurements (eye-tracking, EMG, motion capture) for HCI and XR evaluation.
- Academic writing, publishing, and grant proposal preparation.

TEACHING & SUPERVISION

- Supervision of MSc and BSc theses across computer science, cognitive psychology, and fashion studies.
- Mentoring interdisciplinary student teams and leading collaborative research projects.
- Delivery of lectures and workshops in XR, AI, and HCI.

TECHNICAL & DEVELOPMENT SKILLS

- XR development: Unity, Unreal, ARFoundation, MRTK, SteamVR, OpenXR; hardware including HoloLens, Oculus, HTC Vive.
- Programming: Python (data analysis, ML/DL), C# (Unity), C++, Java, Kotlin, Arduino, HTML.
- Deep Learning: Design, training, and optimization of neural networks (CNNs, RNNs, Transformers) using PyTorch/TensorFlow.
- Large Language Models (LLMs): Hands-on experience implementing Transformer-based architectures, including tokenization, embeddings, attention mechanisms, and training pipelines; application to generative and interactive systems.
- Statistical Analysis: evaluation metrics, and statistical modeling with Python and R.

DESIGN & PROTOTYPING

- UX/UI design and interaction design for XR applications.
- Prototyping with Figma, SketchUp; media editing with Kdenlive and Audacity.

References available upon request.